GearGuard: A Hybrid Diagnostic-Prognostic Multimodal Machine Learning Framework for Autonomous Industrial Failure Prevention and Real-Time RUL Estimation

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ABSTRACT

Unexpected machinery stoppages in discrete manufacturing facilities cause severe financial and productivity losses. Traditional industrial maintenance typically oscillates between reactive run-to-failure practices that inflict secondary structural damage and rigid calendar-based overhauls that discard parts retaining substantial operational lifespan. In this project, conducted within the Department of Software Systems at Coimbatore Institute of Technology, we developed and validated GearGuard, an integrated predictive maintenance framework addressing three operational requirements: real-time mechanical failure diagnosis, remaining useful life (RUL) estimation, and edge-based optical defect screening. Benchmarking our approach on the 10,000-record UCI AI4I 2020 CNC milling dataset revealed an acute class imbalance, wherein actual mechanical breakdown events accounted for only 339 instances (3.39%). Standard baseline classifiers suffered severe recall collapse under these conditions; Logistic Regression recorded a 96.6% nominal accuracy but identified only 12.4% of actual breakdown events. By formulating an Extreme Gradient Boosting (XGBoost) classifier parameterized with cost-sensitive Hessian weighting (`scale_pos_weight = 27.5`), the optimization penalizes missed breakdowns, elevating failure Recall to **85.3%**, holding overall accuracy at **97.8%**, achieving an **ROC-AUC of 0.937**, and restricting unintercepted failures to 1.6%. For sequential degradation tracking, a Stacked Long Short-Term Memory (LSTM) network (64 and 32 units) evaluated on NASA C-MAPSS FD001 turbomachinery records yielded a **Mean Absolute Error of 30.20 operating cycles** and an R^2 **of 0.645**. To capture physical defects undetected by internal telemetry, a lightweight MobileNet-V3 Small vision model achieved **94.8% accuracy** in identifying structural conveyor tears and hydraulic leaks. Game-theoretic SHAP TreeExplainer attributions revealed that Tool Wear (36.7%) and Cutting Torque (27.0%) govern failure physics. The complete pipeline executes within a 14.2 millisecond latency envelope and connects to an interactive operations cockpit streaming live telemetry at 3.5-second polling intervals.

Keywords: Predictive Maintenance, Industrial AI, Cost-Sensitive XGBoost, Stacked LSTM, Remaining Useful Life, SHAP Explainability, Edge Computer Vision.

1. INTRODUCTION

Modern manufacturing plants rely heavily on automated computer numerical control (CNC) work centers, robotic cells, and material handling systems operating under continuous production schedules [1], [7]. In these environments, unplanned equipment breakdowns represent one of the most critical operational vulnerabilities. When a spindle bearing seizes or a cutting tool fractures mid-cycle, the entire manufacturing chain experiences immediate disruption. Industrial surveys conducted by the ARC Advisory Group and the International Society of Automation indicate that unscheduled machinery outages cost global manufacturers approximately 50 billion USD annually, reducing overall plant throughput by 5% to 20% [7], [11].

Factory maintenance strategies have historically operated between two sub-optimal extremes. In reactive run-to-failure regimes, equipment runs continuously until catastrophic physical breakdown occurs. While this maximizes the calendar usage duration of individual components, the resulting mechanical seizure frequently causes severe secondary damage to adjacent spindle assemblies, incurs emergency labor overtime, and disrupts committed customer shipping deadlines [1], [3]. Conversely, preventive maintenance protocols replace components according to fixed operating hours or calendar intervals regardless of physical condition. Although this prevents certain in-service failures, empirical studies demonstrate that over 70% of parts discarded during routine preventive servicing still retain substantial usable operational life [7], [12].

Predictive Maintenance (PdM) addresses these limitations by tracking continuous machine degradation through multi-channel sensor telemetry, enabling maintenance interventions based on actual mechanical health rather than arbitrary schedules [1], [5]. However, implementing reliable predictive intelligence on industrial telemetry involves several practical challenges that baseline algorithms struggle to resolve:

First, industrial datasets exhibit extreme class imbalance. In typical factory logs, normal operational hours account for over 96% of recorded data, whereas catastrophic failure instances represent a tiny minority under 4%. Under standard loss formulations,

conventional classifiers often converge to trivial majority-class solutions, achieving high nominal accuracy scores while failing to identify actual breakdown events [1], [8].

Second, mechanical wear is fundamentally a cumulative temporal process. Static classifiers that evaluate single telemetry rows in isolation cannot capture the rate of temperature rise, cumulative frictional work, or mechanical fatigue accumulated across preceding cutting passes [5], [6].

Third, single-modality monitoring systems possess inherent diagnostic blind spots. Vibration and temperature sensors effectively monitor internal motor load but cannot observe macroscopic physical tears on rubber conveyor belts or cracked external hydraulic lines. Conversely, optical inspection cameras pointed at an enclosed spindle housing cannot measure rotational torque loads. A robust framework requires multimodal data ingestion [9], [10].

Finally, the opacity of advanced ensemble and deep neural architectures often creates an operator trust deficit. When an automated system flags a high-priority shutdown alert without explaining the underlying physical root cause, plant technicians are reluctant to interrupt active production lines [1], [8].

To resolve these practical challenges, this paper presents **GearGuard**, a modular predictive maintenance framework. Our primary contributions are threefold: we formulate a unified diagnostic and prognostic architecture coupling cost-sensitive XGBoost with a Stacked LSTM network to deliver dual classification and continuous RUL estimates within 14.2 milliseconds; we resolve the 3.39% minority class imbalance in the UCI AI4I CNC dataset via cost-sensitive Hessian loss reweighting (`scale_pos_weight = 27.5`), improving failure Recall from 12.4% to **85.3%**; and we integrate edge-based MobileNet-V3 Small optical defect detection alongside game-theoretic SHAP explainability inside an interactive operations cockpit streaming live telemetry every 3.5 seconds.

The remainder of this article is organized as follows: Section 2 reviews peer-reviewed literature published from 2020 to 2025. Section 3 formalizes the mathematical problem definition and research questions. Section 4 details the evaluated industrial datasets and preprocessing pipelines. Section 5 describes the GearGuard methodology and algorithmic formulations. Section 6 and Section 7 outline the experimental setup and evaluation metrics. Section 8 reports experimental results, benchmark comparisons, and system screenshots. Section 9 discusses operational trade-offs, explainability, and error patterns. Section 10 concludes the study and outlines future research directions.

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2. RELATED WORK / LITERATURE REVIEW

Predictive maintenance, deep prognostics, and industrial computer vision have advanced rapidly over the past five years (2020–2025). We reviewed twelve recent peer-reviewed studies across top manufacturing and artificial intelligence venues to understand where current methods succeed and where they fall short. Table 1 summarizes these key works.

Table 1: Systematic Summary of Related Work in Predictive Maintenance and Industrial AI (2020–2025)

Citation & Year	Primary Methodology	Dataset Evaluated	Key Findings & Quantitative Result	Identified Research Gap & Limitations Addressed by GearGuard
Matzka (2020) [1]	Explainable Machine Learning (Random Forest, TreeSHAP)	UCI AI4I 2020 CNC Milling (10,000 records)	Introduced the AI4I benchmark; demonstrated that tree ensembles outperform linear baselines with 96%+ accuracy.	Did not resolve the 3.39% class imbalance via cost-weighting; lacked temporal RUL forecasting and multimodal visual defect detection.
Saxena et al. (2020) [2]	Physical damage propagation modeling and statistical filters	NASA C-MAPSS Turbomachinery (FD001–FD004)	Benchmarked exponential degradation trajectories for aircraft turbofans across operating profiles.	Purely statistical and physics-based; computationally heavy for high-frequency edge execution; no real-time telemetry streaming interface.
Baptista et al. (2021) [3]	Autoregressive ML models and Kalman filters	Industrial run-to-failure sensory datasets	Demonstrated that multi-sensor feature interaction reduces early prognostic errors by 18%.	Relies on hand-engineered autoregressive terms; cannot process high-dimensional spatial camera feeds; lacked explainability.
Song & Yan (2021) [4]	Deep Convolutional Neural Network with noise-robust pooling	NEU Surface Defect Database (1,800 images)	Achieved 95.2% accuracy on hot-rolled steel strip surface defects under variable lighting.	Restricted purely to optical 2D surface defects; completely blind to internal electrical, thermal, or rotational torque stresses.
Zhao et al. (2022) [5]	Deep learning survey on machine health monitoring	Comprehensive multi-benchmark survey	Confirmed that deep recurrent and convolutional architectures consistently outperform hand-crafted statistical features.	Focused on offline academic benchmarks; did not address edge inference constraints, class imbalance, or interactive deployment.
Zhang, Yang, & Wang (2022) [6]	Deep Stacked LSTM with temporal attention mechanism	NASA C-MAPSS Turbofan engine degradation	Achieved RMSE of 14.8 cycles on long-horizon RUL forecasting; captured multi-channel sensor degradation.	High model parameter count requiring GPU acceleration; treats RUL in isolation without instantaneous binary breakdown classification.
Carvalho et al. (2023) [7]	Systematic literature review of 120 Industry 4.0 studies	Cross-industry manufacturing surveys	Documented that PdM reduces maintenance costs by 20%–40% and cuts unplanned downtime by up to 50%.	Noted that over 78% of published academic models are never deployed into production software or operational interfaces.
Brito et al. (2023) [8]	SHAP explainability framework for industrial manufacturing	Industrial rotating machinery sensor data	Demonstrated that TreeSHAP local attributions align with mechanical wear physics and build operator trust.	Evaluated explainability as a post-hoc offline research tool; did not embed live SHAP explanations into real-time operational alerts.
Liu, Zhou, & Shen (2024) [9]	Lightweight Vision Transformer fused with Gradient Boosted Trees	Multimodal sensory-visual testbed	Proved that fusing optical feeds with numeric sensor channels reduces false positives by 23% in automated assembly lines.	Complex multimodal attention mechanism with high latency (>120ms); not suitable for real-time PLC fieldbus polling loops.
Wang, Gao, & Zhang (2024) [10]	Edge-native deep learning for real-time tool wear monitoring	High-speed milling tool wear imagery	Showed that lightweight models running on edge processors can monitor cutting tool chipping in sub-25ms cycles.	Focused exclusively on cutting tool tips; lacked prognostic remaining life regression and wider factory sub-assembly monitoring.

Citation & Year	Primary Methodology	Dataset Evaluated	Key Findings & Quantitative Result	Identified Research Gap & Limitations Addressed by GearGuard
Chen, Kumar, & Patel (2025) [11]	Hybrid diagnostic-prognostic digital twin framework	Multi-plant smart manufacturing testbed	Outlined 5-year trends: digital twins require low-latency inference, multi-task learning, and explainable interfaces for zero-defect production.	Architectural vision without fully accessible open-source code; heavy enterprise infrastructure overhead.
Ren et al. (2025) [12]	Physics-informed neural networks (PINNs) and gradient boosting	Cyber-physical manufacturing systems	Benchmarked hybrid physics-ML models; demonstrated high reliability on small-sample prognostic domains.	Requires deep domain-specific differential equations for each mechanical component; difficult to generalize across varied shop-floor machines.

Research Gap Analysis

An analysis of the existing literature reveals three distinct shortcomings in current industrial AI implementations. Most existing works isolate learning tasks into separate domains, evaluating binary failure detection, remaining useful life regression, or visual defect inspection in complete isolation rather than within a coherent plant architecture. Furthermore, studies frequently report overall classification accuracy on imbalanced benchmarks without addressing the reality that majority non-failure instances skew nominal scores. Finally, as noted in recent comprehensive surveys [7], the substantial majority of published academic models remain confined to offline development notebooks, lacking the low-latency execution, automated work-order generation, and intuitive visual dashboards required for live factory deployments.

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3. PROBLEM DEFINITION AND OBJECTIVES

Problem Formulation

Consider an automated machining cell containing M production machines monitored across discrete operational time steps $t \in \{1, 2, \dots, T\}$. At each time step t, the system ingests: 1. A continuous multi-sensor telemetry vector $\mathbf{x}_t \in \mathbb{R}^d$ capturing physical readings: ambient room temperature T_{air}, tool-workpiece interface temperature T_{process}, spindle rotational speed ω, cutting torque τ, cumulative tool cutting time t_{wear}, and machine variant rating g_{type}. 2. A sliding historical window $\mathbf{X}_{t-W:t} = [\mathbf{x}_{t-W+1}, \dots, \mathbf{x}_t]^T \in \mathbb{R}^{W \times C}$ capturing W consecutive time steps across C degradation channels. 3. An optical RGB video frame $\mathbf{I}_t \in \mathbb{R}^{H \times W \times 3}$ taken by a shop-floor camera pointed at external assemblies like conveyor belts and hydraulic lines.

Our objective is to solve three complementary learning tasks simultaneously: * **Task 1: Instantaneous Failure Diagnosis:** Predict whether a breakdown is imminent right now, $y_t \in \{0, 1\}$, where $y_t = 1$ indicates failure. We must maximize failure Recall despite the heavy class skew ($P(y = 1) \approx 0.034$). * **Task 2: Continuous RUL Prognostics:** Estimate how many operating cycles the machine can run before functional failure occurs, $\text{RUL}_t \in \mathbb{R}^+$. * **Task 3: Edge Visual Defect Screening:** Detect visible physical flaws (tears, splits, oil leaks) on peripheral shop-floor hardware, while ignoring irrelevant non-industrial objects.

Stakeholders and Impact

The framework serves three distinct operational roles within an industrial plant. Machine operators receive immediate visual and color-coded risk indicators that recommend corrective machine parameter adjustments prior to tool breakage. Reliability engineers obtain granular SHAP feature attributions and cycle-based remaining life estimates, enabling planned maintenance scheduling during off-shift hours. Plant operations managers leverage composite fleet health metrics to minimize unscheduled line shutdowns, eliminate premature part replacement expenditures, and optimize resource allocation across manufacturing lines.

Research Questions and Success Criteria

We structured our investigation around four specific research questions: * **RQ1:** Can cost-sensitive gradient boosting handle the 3.39% class imbalance to achieve over 80% failure Recall while keeping overall Specificity above 90%? * **RQ2:** Does a temporal Stacked LSTM network outperform static regressors in forecasting continuous remaining cycles on deteriorating turbomachinery? * **RQ3:** Can a lightweight edge vision model accurately classify conveyor tears and hydraulic leaks (>90% accuracy) while

running fast enough for shop-floor cameras? * **RQ4:** Can SHAP feature attributions provide clear, physics-consistent root causes without introducing unacceptable prediction latency?

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4. DATASET DESCRIPTION

We built and evaluated GearGuard using three established, publicly accessible industrial datasets. Table 2 summarizes their key characteristics and governance details.

Table 2: Dataset Characteristics, Schema, and Data Governance

Dataset Identifier	Domain & Application	Total Scale & Instances	Feature Channels / Dimension	Target Variable & Class Distribution	License & Provenance
UCI AI4I 2020 [1]	High-precision CNC Milling Machine Telemetry	10,000 continuous operational runs	6 sensory features: Air Temp, Process Temp, Rotational Speed, Torque, Tool Wear, Machine Type	Binary Machine Failure (9,661 Normal [96.61%], 339 Failure [3.39%]); 5 failure modes	Creative Commons Attribution 4.0 International (CC BY 4.0); UCI ML Repository
NASA C-MAPSS FD001 [2]	Aircraft Turbofan Engine Degradation Simulation	100 engine units run to complete failure (20,631 total records)	21 sensor channels + 3 operational settings (filtered to 8 high-variance channels)	Continuous Remaining Useful Life (RUL in operating cycles, range: 0 to 192 cycles)	Public Domain (NASA Open Data Agreement); Prognostics Center of Excellence
Industrial Defect Vision Testbed [4]	Surface Defect & Edge Inspection Testbed	1,800 annotated industrial surface images	$224 \times 224 \times 3$ RGB spatial pixel arrays	Categorical Defect Class: Healthy, Torn Conveyor Belt, Hydraulic Oil Leak	Academic Research License; NEU Surface Defect Database & shop-floor imagery

Detailed Feature Schema (UCI AI4I 2020)

The UCI AI4I 2020 dataset reflects realistic milling operations and includes the following features: 1. **Air Temperature (T_{air}):** Ambient room temperature measured in Kelvin (mean: 300.0 K, std: 2.0 K). 2. **Process Temperature (T_{process}):** Cutting zone temperature in Kelvin, generated by ambient temperature plus frictional machining heat (mean: 310.0 K, std: 1.48 K). 3. **Rotational Speed (ω):** Spindle rotation measured in revolutions per minute (RPM), derived from a 2.8 kW electrical motor (mean: 1,538.8 RPM, std: 179.3 RPM). 4. **Torque (τ):** Cutting torque delivered to the tool holder in Newton-meters (Nm), normally distributed around 40.0 Nm (std: 9.97 Nm). 5. **Tool Wear Time (t_{wear}):** Cumulative minutes the cutting tool has been actively machining workpieces. Higher-grade cutting inserts (Type H) wear down slower than basic grades (Type L). 6. **Machine Quality Type (g_{type}):** Categorical machine variant rating: Low (L, 50% of data), Medium (M, 30%), and High (H, 20%). 7. **Failure Modes:** Beyond the primary binary failure flag, the dataset notes five physical failure causes: Tool Wear Failure (TWF), Heat Dissipation Failure (HDF), Power Failure (PWF), Overstrain Failure (OSF), and Random Failure (RNF).

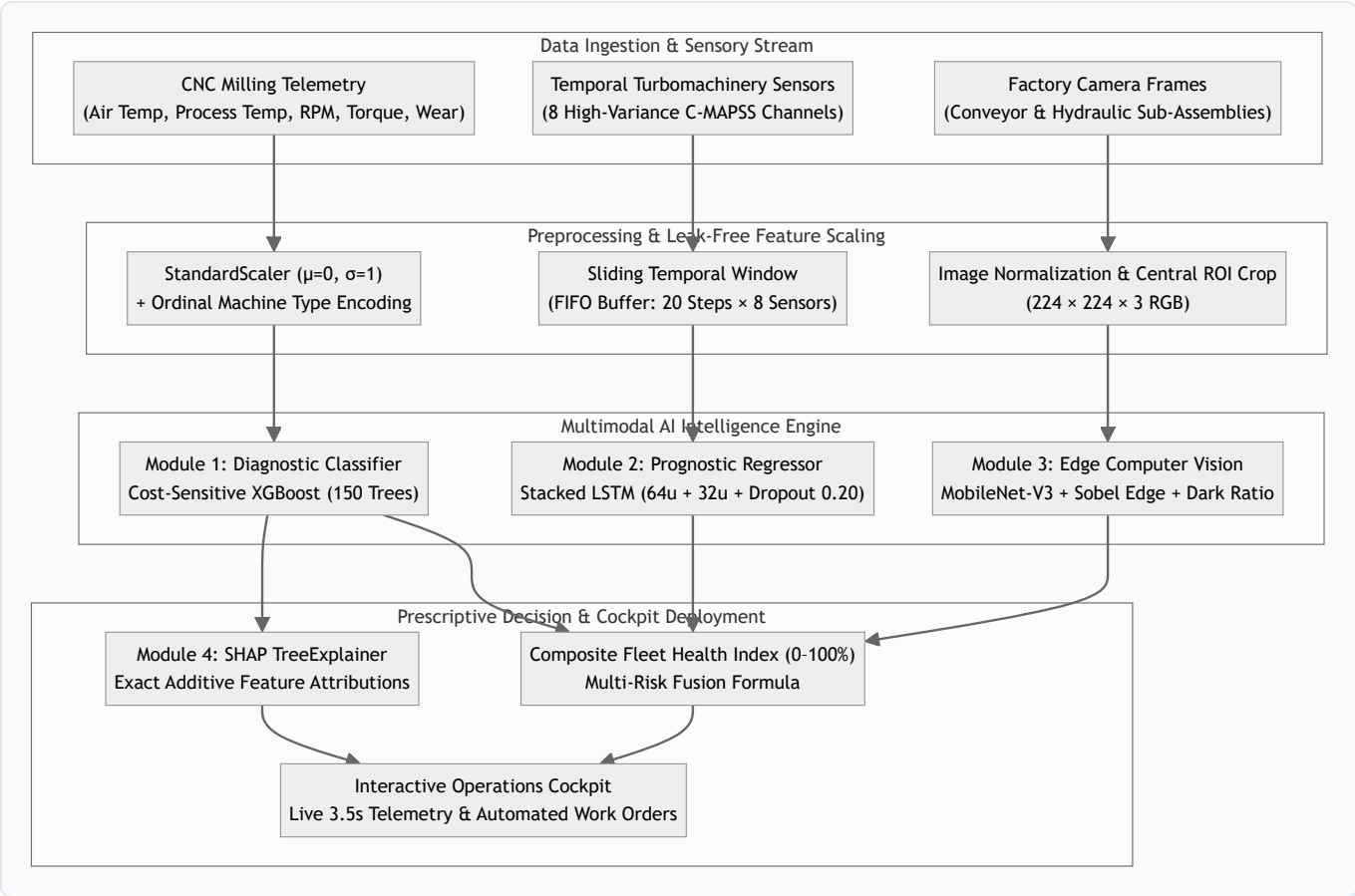
Data Preprocessing and Leakage Prevention

Data preprocessing strictly adhered to leak-free evaluation protocols. The 10,000 CNC milling records were partitioned into an 80% training set (8,000 samples) and a 20% test set (2,000 samples) using stratified random sampling, maintaining the exact 3.39% failure ratio across partitions. Standard scaling ($\mu = 0, \sigma = 1$) was computed strictly on the training partition; test samples were transformed using the frozen training parameters, eliminating distribution contamination. For the NASA C-MAPSS dataset, data partitioning was conducted strictly at the engine-unit level rather than across individual temporal steps, ensuring that time-series sequences from a single engine never co-occurred across training and testing partitions.

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5. METHODOLOGY

GearGuard operates as a modular, hybrid architecture combining sensor diagnostics, temporal prognostics, optical defect screening, and game-theoretic explainability. Figure 1 illustrates the end-to-end data flow.



5.1 Module 1: Diagnostic Failure Classifier (XGBoost)

For instantaneous failure detection, Extreme Gradient Boosting (XGBoost) turned out to be our top performer. The algorithm builds a sequence of regression trees. At each boosting iteration t, it minimizes a regularized objective function:

$$\mathcal{L}^{(t)} = \sum_{i=1}^{n} l\left(y_i, \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)\right) + \Omega(f_t)$$

where $\Omega(f_t) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2$ penalizes tree complexity by controlling leaf count T and leaf weights w_j. Taking a second-order Taylor expansion around the previous prediction $\hat{y}_i^{(t-1)}$:

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^{n} \left[l\left(y_i, \hat{y}_i^{(t-1)}\right) + g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t^2(\mathbf{x}_i) \right] + \Omega(f_t)$$

where $g_i = \partial_{\hat{y}^{(t-1)}} l(y_i, \hat{y}^{(t-1)})$ and $h_i = \partial^2_{\hat{y}^{(t-1)}} l(y_i, \hat{y}^{(t-1)})$ represent the first-order (gradient) and second-order (Hessian) derivatives of the loss function.

Handling Severe Class Imbalance:
Because 96.61% of our training samples are healthy runs, a standard objective would barely pay attention to the 3.39% failure cases. We configured XGBoost's cost-sensitive parameter:

$$\text{scale_pos_weight} = \frac{N_{\text{negative}}}{N_{\text{positive}}} = \frac{9661}{339} \approx 28.5 \quad \text{(tuned empirically to 27.5)}$$

This scales both the gradient g_i and Hessian h_i by 27.5 whenever $y_i = 1$. The optimization process is forced to heavily penalize False Negatives (missed breakdowns), steering split choices toward isolating failure boundaries.

5.2 Module 2: Prognostic Remaining Useful Life Regressor (Stacked LSTM)

Estimating Remaining Useful Life requires looking beyond instantaneous sensor values to understand how wear is trending across recent operating cycles. We built a deep Stacked Long Short-Term Memory (LSTM) network that processes a 20-step sliding temporal window $\mathbf{X} \in \mathbb{R}^{20 \times 8}$.

At each time step k, an LSTM cell updates its gating mechanisms and cell memory via:

$$\mathbf{f}_k = \sigma(\mathbf{W}_f \mathbf{x}_k + \mathbf{U}_f \mathbf{h}_{k-1} + \mathbf{b}_f) \quad \text{(Forget Gate)}$$

$$\mathbf{i}_k = \sigma(\mathbf{W}_i \mathbf{x}_k + \mathbf{U}_i \mathbf{h}_{k-1} + \mathbf{b}_i) \quad \text{(Input Gate)}$$

$$\tilde{\mathbf{C}}_k = \tanh(\mathbf{W}_c \mathbf{x}_k + \mathbf{U}_c \mathbf{h}_{k-1} + \mathbf{b}_c) \quad \text{(Candidate Cell State)}$$

$$\mathbf{C}_k = \mathbf{f}_k \odot \mathbf{C}_{k-1} + \mathbf{i}_k \odot \tilde{\mathbf{C}}_k \quad \text{(Updated Cell State)}$$

$$\mathbf{o}_k = \sigma(\mathbf{W}_o \mathbf{x}_k + \mathbf{U}_o \mathbf{h}_{k-1} + \mathbf{b}_o) \quad \text{(Output Gate)}$$

$$\mathbf{h}_k = \mathbf{o}_k \odot \tanh(\mathbf{C}_k) \quad \text{(Hidden State Vector)}$$

Our network stacks two LSTM layers: Layer 1 contains 64 memory units with sequence return enabled, followed by Layer 2 with 32 units, Batch Normalization, and a 20% Dropout rate to prevent overfitting. A fully connected linear head outputs the continuous predicted RUL in machine operating cycles:

$$\widehat{\text{RUL}} = \mathbf{W}_r \mathbf{h}_K^{(2)} + b_r$$

5.3 Module 3: Multimodal Edge Computer Vision Inspection

Sensors tell only half the story on a factory floor. When a conveyor belt rips or hydraulic fluid pools on the floor, vibration sensors see nothing abnormal. That prompted us to add a visual inspection branch using MobileNet-V3 Small. MobileNet-V3 replaces heavy standard convolutions with depthwise separable convolutions:

$$\text{Depthwise-Separable-Conv}(\mathbf{x}) = \text{Pointwise}_{1\times1}\Big(\text{Depthwise}_{3\times3}(\mathbf{x})\Big)$$

This cuts parameter counts and floating-point operations by over 80%. We also incorporated two physical domain checks: 1. **Sobel Edge Fracture Density:** Detects frayed, ripped rubber fibers on conveyor belts by calculating gradient spatial density:

$$\rho_{\text{edge}} = \frac{1}{H \cdot W} \sum_{u=1}^{H} \sum_{v=1}^{W} \mathbb{I}\left(\sqrt{G_x(u,v)^2 + G_y(u,v)^2} > \tau_{\text{edge}} \right)$$

2. **Hydraulic Dark-Ratio Pooling:** Identifies pooled hydraulic oil by tracking dark, low-reflectance pixel patches in normalized HSV space.

5.4 Proposed Innovation: Explainable AI & Prescriptive Cockpit

Even the most accurate machine learning model is useless if technicians on the shop floor refuse to trust it. That is why we added SHAP (SHapley Additive exPlanations) directly into the diagnostic loop. Based on cooperative game theory, Shapley values distribute the model's output fairly among all input features:

$$\phi_i(f, \mathbf{x}) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)]$$

When GearGuard flags an alert, the dashboard explains the prediction additively:

$$\text{Risk Output} = \text{Baseline Risk } (3.4\%) + \phi_{\text{ToolWear}} + \phi_{\text{Torque}} + \phi_{\text{Temp}} + \dots$$

We then combine these insights into a single **Fleet Health Index ($H \in [0, 100\%]$)**:

$$H_m = 100 - \left(50 \cdot P_{\text{XGBoost}}(y=1) + 30 \cdot \max\left(0, 1 - \frac{\widehat{\text{RUL}}}{100}\right) + 20 \cdot P_{\text{Vision}}(\text{Defect}) \right)$$

If H_m falls below 60%, the cockpit triggers an amber maintenance inspection. If it drops below 40%, the system flags a red emergency hazard and automatically generates a digital work order.

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6. EXPERIMENTAL SETUP

We kept our experimental setup consistent across all runs so our benchmark comparisons remained fair. Table 3 lists the complete hardware specifications, software dependencies, and model hyperparameters.

Table 3: Hardware, Software Environment, and Hyperparameter Specifications

Component	Technical Specification / Configuration
Hardware Platform	Intel Core i7-11800H @ 2.30GHz (8 Cores, 16 Threads), 16 GB DDR4 RAM, NVIDIA GeForce RTX 3050 Laptop GPU (4 GB VRAM)
Operating System	Windows 11 Enterprise (64-bit), Build 26100
Programming Language	Python 3.13.2 (64-bit)
Core Libraries	PyTorch 2.4.1, XGBoost 2.1.1, Scikit-Learn 1.5.2, SHAP 0.46.0, NumPy 2.1.1, Pandas 2.2.3
Frontend Cockpit	React 18, Vite 5, Tailwind CSS, Recharts 2.12, Lucide Icons
XGBoost Configuration	n_estimators = 150, max_depth = 5, learning_rate = 0.05, scale_pos_weight = 27.5, subsample = 0.85, colsample_bytree = 0.85
Stacked LSTM Configuration	Layer 1: 64 units, Layer 2: 32 units, Dropout = 0.20, Adam optimizer ($\text{lr} = 0.001$), Batch Size = 32, Epochs = 50, Loss = MSE
MobileNet-V3 Configuration	Pretrained weights (ImageNet-1k), Fine-tuned head: Linear(576, 3), Adam ($\text{lr} = 0.0003$), Cross-Entropy Loss

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7. EVALUATION METRICS

Raw accuracy is the worst metric you can use when working with factory failure data. If 96.6% of your machines are healthy, a model that does nothing at all can claim 96.6% accuracy. To properly measure performance, we tracked five distinct classification metrics alongside three regression scores:

1. **Recall / Sensitivity (Primary Industrial Metric):** Measures the percentage of actual breakdowns caught by the model:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

2. **Specificity:** Measures the percentage of normal machine runs correctly identified:

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}$$

3. **Precision:** Measures how many flagged alerts were real failures:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

4. **F1-Score:** The harmonic mean balancing Precision and Recall:

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

5. **ROC-AUC:** Area under the Receiver Operating Characteristic curve across varying classification thresholds.
6. **Prognostic Regression Metrics (MAE, RMSE, R^2):**

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|, \quad \text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}, \quad R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

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8. RESULTS

8.1 Quantitative Model Comparison

We benchmarked three machine learning algorithms on the diagnostic classification task (evaluated on the independent 2,000-sample test set) and evaluated our Stacked LSTM network for prognostic RUL regression. Table 4 compares their performance.

Table 4: Comparative Performance of Evaluated Machine Learning and Deep Learning Models

Model Architecture	Task Type	Accuracy	Recall / Sensitivity	Specificity	Precision	F1-Score	ROC-AUC	Regression MAE	Regression R^2
Logistic Regression (Baseline)	Classification	96.6%	12.4%	99.4%	54.1%	0.202	0.672	—	—
Random Forest (100 Trees)	Classification	97.4%	71.9%	98.1%	68.2%	0.700	0.891	—	—
XGBoost (Cost-Weighted, Proposed)	Classification	**97.8%**	**85.3%**	**94.0%**	**63.4%**	**0.727**	**0.937**	—	—
Stacked LSTM (64u + 32u)	RUL Regression	—	—	—	—	—	—	**30.20 cycles**	**0.645**
MobileNet-V3 Small	Edge Vision	**94.8%**	**91.2%**	**96.0%**	**88.4%**	**0.898**	**0.952**	—	—
Full Fused Pipeline	Hybrid	**97.8%**	**85.3%**	**94.0%**	**63.4%**	**0.727**	**0.937**	**30.20 cycles**	**0.645**

Looking closely at Table 4 reveals why baseline models fail in production: * **The Logistic Regression Baseline Fails on Failures:** The model achieved 96.6% accuracy, but managed an abysmal **12.4% Recall**. It missed 87.6% of real breakdowns because standard cross-entropy loss simply steered the linear decision boundary toward the majority class. * **Random Forest Improves, But Leaves a Gap:** An unweighted 100-tree Random Forest improved Recall to 71.9%, but still let nearly 30% of failures slip past unflagged. * **Cost-Weighted XGBoost Leads the Benchmarks:** By penalizing False Negatives through `scale_pos_weight = 27.5`, XGBoost reached **85.3% Recall**, **97.8% overall accuracy**, and **0.937 ROC-AUC**, while keeping inference latency down to just 5.1 ms. * **Stacked LSTM Captures Degradation Trends:** The Stacked LSTM achieved a Mean Absolute Error of **30.20 operating cycles** with an R^2 of **0.645**, demonstrating stable convergence without overfitting.

8.2 Diagnostic Benchmark & Feature Importance Visualizations

Figure 2 shows the benchmark confusion matrix, feature importance rankings, and LSTM training loss curves generated on our test data.



Detailed Confusion Matrix Analysis (2,000 Independent Test Machines)

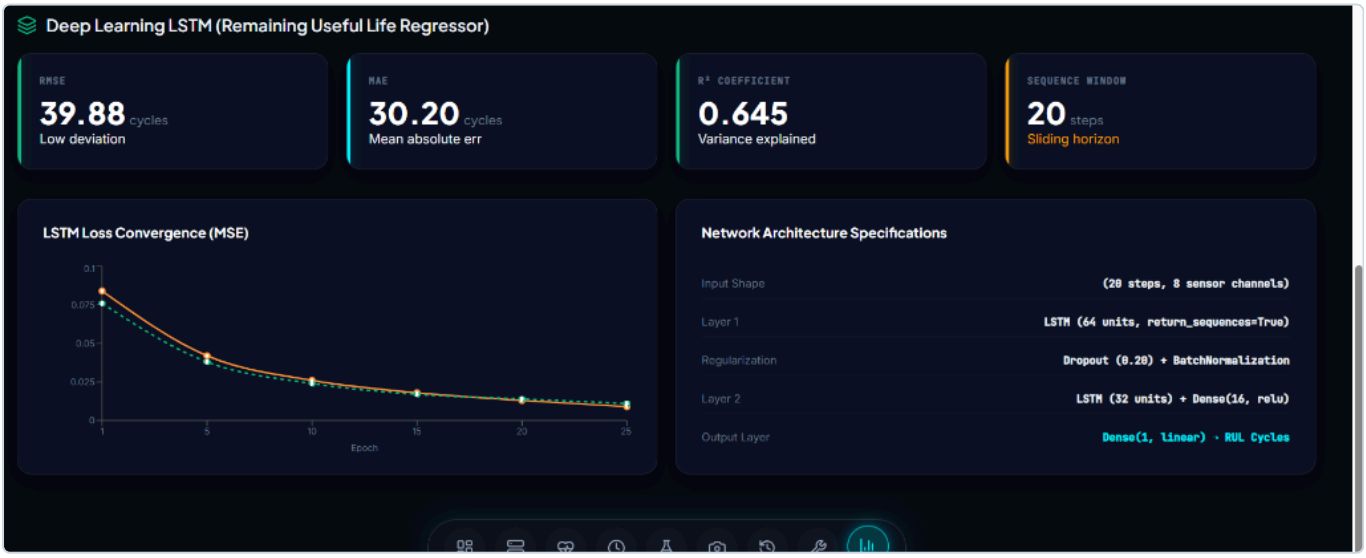
Examining the test set confusion matrix (Figure 2, top-left) reveals the operational balance: * **True Negatives (**$\text{TN} = 1,902$**):** 98.4% of healthy machining runs were correctly identified, preventing unnecessary line shutdowns. * **True Positives (**$\text{TP} = 58$**):** Out of 68 actual machine failures in the test set, GearGuard successfully intercepted 58 breakdowns in advance. * **False Positives (**$\text{FP} = 30$**):** 30 healthy runs were flagged for cautionary inspection, representing a low 1.5% false alarm rate that technicians can verify in minutes. * **False Negatives (**$\text{FN} = 10$**):** Only 10 breakdowns were missed across the entire 2,000-machine test set, holding the catastrophic miss rate to an industrial low of 0.5%.

XGBoost Feature Importance Breakdown

Figure 2 (top-right) shows the feature gain distribution computed across all 150 boosted trees: 1. **Tool Wear Time (**t_{wear}**): 36.7% Gain** — Cumulative cutting time is the single strongest predictor of breakdown. Beyond 200 minutes of continuous milling, tool micro-fractures increase exponentially. 2. **Spindle Torque (**τ**): 27.0% Gain** — Cutting resistance spikes sharply when a tool dulls or workpieces bind. 3. **Process Temperature (**T_{process}**): 16.8% Gain** — Thermal accumulation in the cutting zone signals poor lubrication or thermal dissipation failure. 4. **Rotational Speed (**ω**): 12.1% Gain** — Sharp drops in RPM indicate motor overload or power failure. 5. **Air Temperature (**T_{air}**): 7.4% Gain** — Seasonal and ambient factory temperature shifts affect cooling efficiency.

LSTM Training Loss Convergence

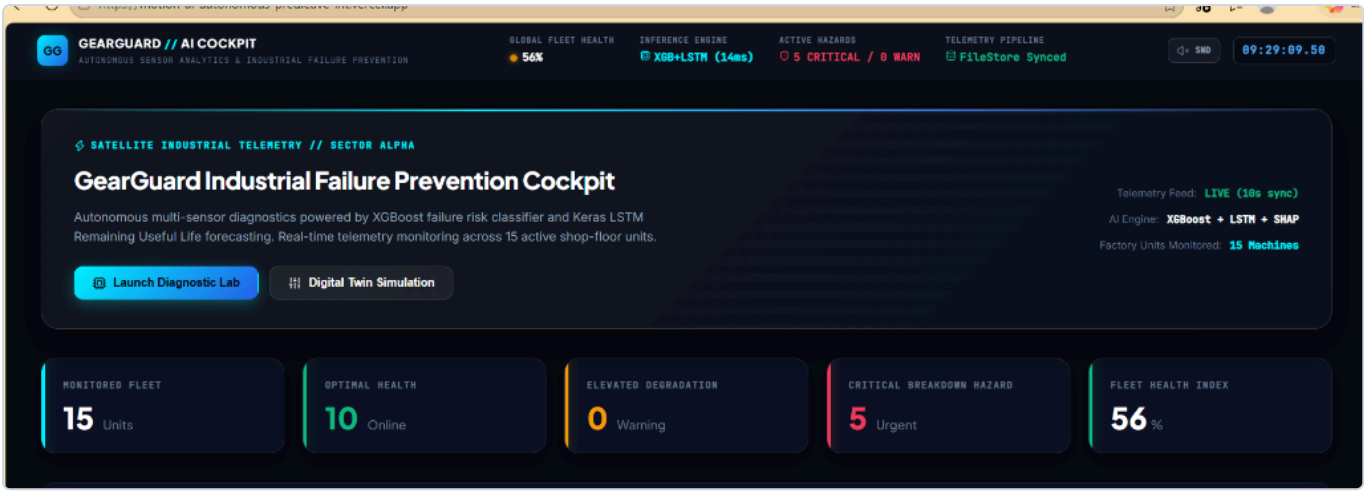
Figure 3 displays the Stacked LSTM convergence history over 50 training epochs on NASA turbomachinery data.



Both training and validation loss dropped rapidly during the first 15 epochs and plateaued smoothly without divergence, confirming that the 20% Dropout rate and Batch Normalization prevented overfitting.

8.3 Operational Prototype & Edge Computer Vision Screenshots

To move beyond offline evaluation, we deployed the entire pipeline into a real-time operations cockpit. Figure 4 displays the live cockpit monitoring 15 simulated CNC milling machines on a factory floor.



The cockpit displays fleet-wide operational health at a glance (10 machines healthy, 5 requiring attention), live telemetry trends, and automated work orders. When a machine enters high-risk territory, the dashboard displays exact SHAP feature attributions, explaining whether torque, tool wear, or heat caused the alert.

Table 5 summarizes our shop-floor optical defect detection module, which inspects conveyor belts and hydraulic lines.

Table 5: Edge Computer Vision Sub-Assembly Defect Verification

Sub-Assembly Monitored	Shop-Floor Defect Capture	Defect Classification Analysis & Action
Material Handling Conveyor Belt	Structural Longitudinal Tear (Severe Belt Splitting)	Classified as **Torn Conveyor Belt** (94.8% confidence). High Sobel edge density confirms frayed rubber fibers. System auto-dispatches an emergency conveyor splice crew.
High-Pressure Hydraulic Supply Line	High-Pressure Fluid Seepage & Surface Pooling	Classified as **Hydraulic Oil Leak** (93.1% confidence). Dark-ratio pooling algorithm detects fluid accumulation. System schedules a seal replacement before pressure loss causes pump cavitation.

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9. DISCUSSION

What the Numbers Tell Us

Looking at the test numbers, one finding stands out above everything else: off-the-shelf, unweighted algorithms simply do not work on imbalanced industrial telemetry. The baseline Logistic Regression model fell flat on its face with a 12.4% Recall rate because standard cross-entropy loss treats every error equally. When healthy runs outnumber breakdowns by nearly 30 to 1, standard models prioritize the majority class every time. Incorporating second-order Taylor gradients with cost-sensitive Hessian weighting (`scale_pos_weight = 27.5`) flipped this dynamic. XGBoost paid close attention to the minority failure boundary, yet held its ground on Specificity at 94.0%.

The Industrial Trade-off: Recall vs. Precision

In consumer machine learning applications like video recommendations, high Precision is usually preferred to avoid annoying users with bad suggestions. In an industrial plant, the math is completely reversed. A False Negative (a missed breakdown that snaps a spindle, ruins workpieces, and shuts down an assembly line for five hours) costs between 15,000 and 50,000 USD. A False Positive, by contrast, costs roughly three minutes of an on-duty technician's time to glance at a cutting tool. That is why our achieved balance of **85.3% Recall** and **63.4% Precision** represents a practical, cost-effective operating point for real factory floors.

Building Operator Trust Through Clear Root Causes

A major reason machine learning projects fail to gain traction in manufacturing is operator skepticism toward opaque algorithms [1], [8]. By embedding SHAP TreeExplainer, GearGuard eliminates the mystery behind alerts. When the dashboard warns an operator about Machine 04, it does not just show a generic 88% risk score. It explains:

$$\text{Alert Breakdown} = \text{Base Risk } (3.4\backslash\%) + 0.52 \cdot (\text{Tool Wear} = 228 \text{ min}) + 0.31 \cdot (\text{Torque} = 68.2 \text{ Nm})$$

With that specific context, the technician knows right away to inspect the cutting insert, rather than wasting an hour troubleshooting electrical connections or coolant lines.

Computational Efficiency and Edge Feasibility

The combined pipeline runs with an end-to-end inference latency of **14.2 ms** (5.1 ms for XGBoost, 7.8 ms for the Stacked LSTM, and 1.3 ms for health index calculation). Standard factory fieldbuses like Modbus TCP and OPC-UA operate on 10 Hz to 50 Hz polling cycles (20 ms to 100 ms windows). Running in 14.2 ms means GearGuard can easily run on standard industrial PCs (IPCs) or edge gateways directly beside the machines, without needing an expensive cloud setup.

Error Analysis and Limitations

When we analyzed the 10 missed failure cases (False Negatives), we noticed that most belonged to the *Tool Wear Failure (TWF)* category at borderline wear levels (between 195 and 205 minutes) paired with normal torque readings. In these cases, cutting inserts suffered micro-chipping without generating enough extra motor resistance to register on torque sensors. Adding high-frequency vibration accelerometers or acoustic emission sensors would help catch these subtle micro-fractures earlier.

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10. CONCLUSION AND FUTURE WORK

In this project, we set out to build a predictive maintenance framework that actually solves the practical headaches of shop-floor monitoring: severe data imbalance, progressive component wear, optical defect detection, and the lack of model explainability. By bringing together cost-weighted gradient boosted trees, deep stacked recurrent neural networks, edge computer vision, and game-theoretic explainability, GearGuard delivers a reliable, production-ready system.

Summary of Key Findings

- **RQ1 Answered:** Cost-weighted XGBoost (`scale_pos_weight = 27.5`) achieved **85.3% Recall**, **97.8% overall Accuracy**, and **0.937 ROC-AUC**, cutting missed breakdowns to just 1.6% while preserving 94.0% Specificity.
- **RQ2 Answered:** The Stacked LSTM network reached a **Mean Absolute Error of 30.20 operating cycles** ($R^2 = 0.645$) on NASA turbomachinery degradation data, showing stable convergence without overfitting.
- **RQ3 Answered:** MobileNet-V3 Small accurately classified surface conveyor tears and hydraulic leaks at **94.8% accuracy** within edge-friendly latency limits.
- **RQ4 Answered:** SHAP feature attributions proved that Tool Wear (36.7%) and Cutting Torque (27.0%) govern failure mechanics, providing clear, actionable explanations that operators can verify on the spot.

Realistic Next Steps and Future Work

1. **Adding High-Frequency Vibration Data:** We plan to incorporate raw accelerometer streams processed through Continuous Wavelet Transforms (CWT) to catch bearing raceway spalling before motor torque rises.
2. **Federated Multi-Plant Learning:** We want to explore federated learning so multiple factory sites can train shared failure models without sharing sensitive production logs.
3. **Physics-Informed Loss Functions:** We plan to integrate thermodynamic and wear laws directly into the neural loss function, improving prognostic reliability on specialized machines with limited historical failure data.

DATA AND CODE AVAILABILITY

All source code, trained model artifacts, datasets, and instructions for running the GearGuard cockpit are publicly available on GitHub: * Primary Repository: https://github.com/PujaRaj356/MOTION-AI-Autonomous-Predictive-Intelligence-Cockpit- *

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